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Course:CSA1718-Ai

Experiment 14: Decision Tree Classification

Aim

To implement the Decision Tree Classification algorithm using Python and classify data based on the given features.

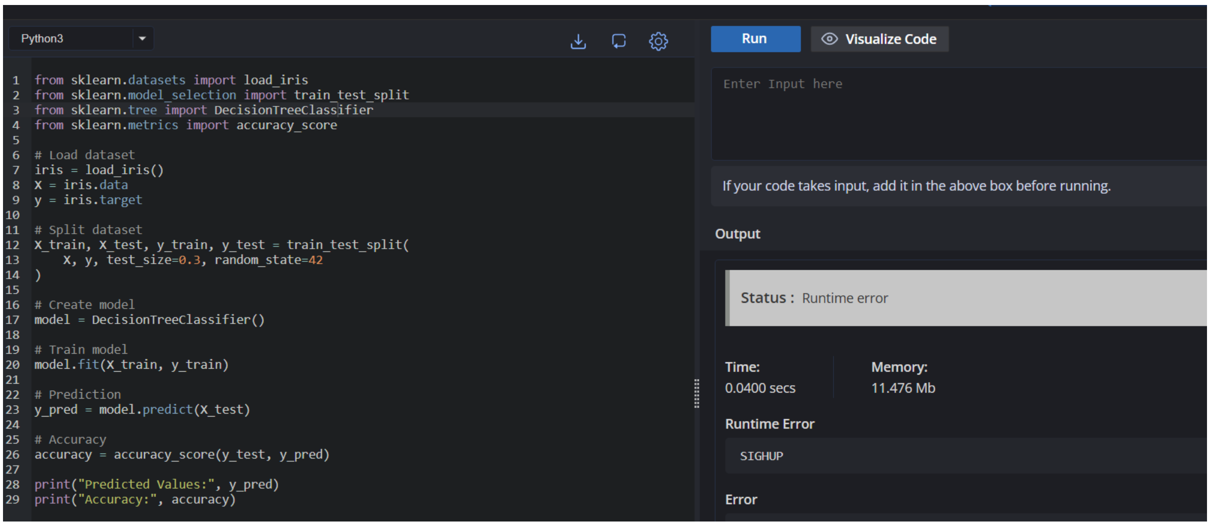
Procedure

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create a Decision Tree Classifier.
5. Train the model using the training data.
6. Predict the output for the test data.
7. Calculate the accuracy of the model.
8. Display the results.

Python Code

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score


# Load dataset

iris = load_iris()

X = iris.data

y = iris.target
```

```
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create model
model = DecisionTreeClassifier()

# Train model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Predicted Values:", y_pred)
print("Accuracy:", accuracy)
```



Result

The Decision Tree Classification model was implemented successfully, and the predicted values were obtained with high classification accuracy.

Outcome

The experiment demonstrated how a Decision Tree can classify data efficiently using supervised machine learning.

Experiment 15: Feed Forward Neural Network (FFNN)

Aim

To implement a Feed Forward Neural Network using Python for classification.

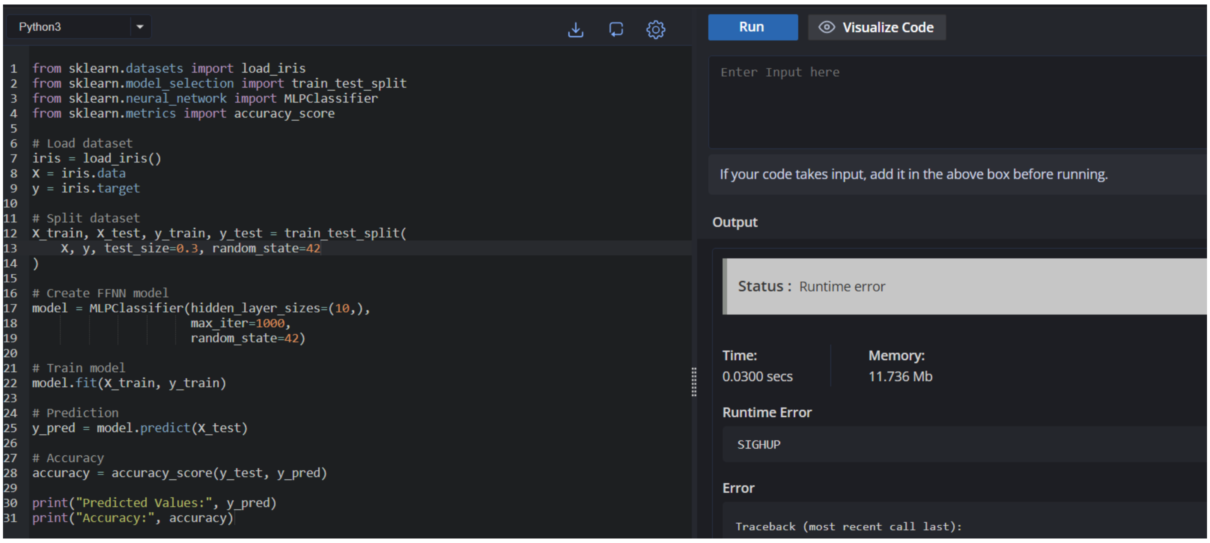
Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing data.
4. Create a Feed Forward Neural Network model.
5. Train the model.
6. Predict the test data.
7. Evaluate the model accuracy.
8. Display the results.

Python Code

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split
```

```
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create FFNN model
model = MLPClassifier(hidden_layer_sizes=(10,),
                      max_iter=1000,
                      random_state=42)

# Train model
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Predicted Values:", y_pred)
print("Accuracy:", accuracy)
```



Result

The Feed Forward Neural Network was successfully trained and tested, producing accurate predictions on the test dataset.

Outcome

The experiment demonstrated the implementation of a Feed Forward Neural Network for solving a classification problem using Python.

These sections are ready to copy into a **Microsoft Word (.docx)** lab record.